INSTITUTO COLLEGE

# MATH 110: General Education Math

Fall 2026

# Sets Review

*Name:* _____

## Sets and Elements

A **set** is a collection of distinct objects, called its **elements**. We list the elements of a set between braces, for example $A = \{\text{yellow}, -4, \text{raincoat}\}$ denotes a set with $3$ elements – the elements are yellow, $-4$, and raincoat. We write $x \in A$ when $x$ is an element of $A$, and $x \notin A$ when it is not. The order of the elements does not matter, and each element is listed only once. The **empty set**, denoted $\emptyset$, is the unique set that contains no elements.

1. Let $R = \{\text{red}, \text{green}, \text{blue}\}$ and $N = \{2, 7, 9\}$. Write *true* or *false*.

   (a) green $\in R$ _____

   (b) yellow $\in R$ _____

   (c) yellow $\notin R$ _____

   (d) $7 \in N$ _____

   (e) $3 \in N$ _____

   (f) red $\in N$ _____

   (g) $\{2, 7, 9\} = \{9, 2, 7\}$ _____

2. Write each set by listing its elements between braces.

   (a) $W$, the vowels in the word NURSE. $W =$ _____

   (b) $T$, the even numbers between $1$ and $9$. $T =$ _____

   (c) $J$, the months of the year whose names begin with the letter J.
   $J =$ _____

   (d) $Z$, the whole numbers that are greater than $5$ and less than $6$.
   $Z =$ _____

## Cardinality

The **cardinality** of a set $A$, written $|A|$, is the number of elements in $A$. A set is a collection of objects, but its cardinality is a number. The empty set, $\emptyset$, is the unique set with no elements, so $|\emptyset| = 0$.

3. Find the cardinality of each set.

   (a) $\big|\{\text{red}, \text{green}, \text{blue}\}\big| =$ _____

   (b) $\big|\{2, 7, 9\}\big| =$ _____

   (c) $\big|\{100\}\big| =$ _____

   (d) $|\emptyset| =$ _____

   (e) $\big|\{0\}\big| =$ _____

   (f) $\big|\{\text{Mon}, \text{Wed}\}\big| =$ _____

4. Let $N = \{2, 7, 9\}$.

   (a) Find $|N|$. $|N| =$ _____

   (b) Is $|N|$ an element of $N$? Explain.

   (c) A student writes $N = 3$. What is wrong with this? What should the student write instead?

5. Each set below is described in words. Find its cardinality without listing all of its elements.

(a) $E$, the set of even numbers between $1$ and $100$, including $100$. $|E| =$ _____

(b) $W$, the set of whole numbers from $1$ to $250$. $|W| =$ _____

(c) $T$, the set of whole numbers from $10$ to $30$. $|T| =$ _____

(d) $M$, the set of multiples of $5$ from $5$ to $100$. $|M| =$ _____

(e) $O$, the set of odd numbers between $0$ and $20$. $|O| =$ _____

(f) $D$, the set of days in the month of November. $|D| =$ _____

## Set or Number?

6. For each expression, write *set* or *number*.

| Expression | Set or number? |
|---|---|
| $\{4\}$ | |
| $4$ | |
| $\lvert\{4\}\rvert$ | |
| $\emptyset$ | |
| $\lvert\emptyset\rvert$ | |
| $\{\text{Ana}, \text{Ben}\}$ | |
| $\lvert\{\text{Ana}, \text{Ben}\}\rvert$ | |
| $0$ | |

## Union and Intersection

The **union** of two sets $A$ and $B$, denoted $A \cup B$, is the set of elements that are in $A$ or in $B$ (or in both). The **intersection** of $A$ and $B$, denoted $A \cap B$, is the set of elements that are in both $A$ and $B$. The union and the intersection of two sets are sets, so when we write their elements, we use braces. Often all of the sets we are working with are made up of elements from one larger set, called the **universal set** $U$, which consists of all the elements we are working with.

**The clinic.** Many of the problems below are about one clinic. Eight patients visit the clinic today: Ana, Ben, Carla, David, Eva, Hugo, Iris, and Juan. Using their first initials, the universal set is

$$U = \{\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{D}, \mathrm{E}, \mathrm{H}, \mathrm{I}, \mathrm{J}\}.$$

The patients who have a fever form the set $F = \{\mathrm{A}, \mathrm{C}, \mathrm{D}, \mathrm{I}\}$, and the patients who have had a flu shot this year form the set $V = \{\mathrm{B}, \mathrm{C}, \mathrm{E}, \mathrm{I}, \mathrm{J}\}$.

7. Use the clinic sets $U$, $F$, and $V$.

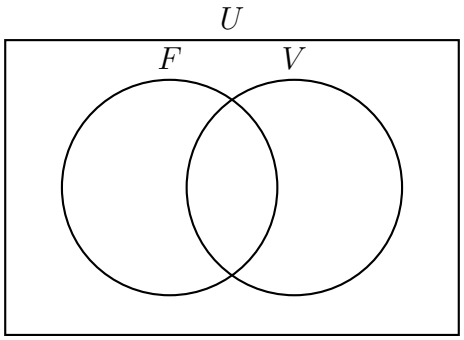
   (a) Find $F \cup V$. $F \cup V =$ _____

   (b) Find $F \cap V$. $F \cap V =$ _____

   (c) Find each cardinality.

   $|F| =$ _____ $|V| =$ _____ $|F \cup V| =$ _____ $|F \cap V| =$ _____

8. Use the clinic sets $U$, $F$, and $V$. Write the initial of each of the $8$ patients in the correct region of the Venn diagram.




   (a) Which patients are in $F$ but not in $V$? _____

   (b) Which patients are in both $F$ and $V$? _____

   (c) Which patients are in neither $F$ nor $V$? _____

## Subsets

A set $A$ is a **subset** of a set $B$, written $A \subseteq B$, if every element of $A$ is also an element of $B$. Every set is a subset of itself, so $A \subseteq A$, and the empty set is a subset of every set, so $\emptyset \subseteq A$.

9. Use the clinic sets $U$, $F$, and $V$, and your answers to Problems 7 and 8. Write *true* or *false*. If a statement is false, name an element that shows it is false.

(a) $F \subseteq F \cup V$ _____

(b) $V \subseteq F \cup V$ _____

(c) $F \cap V \subseteq F$ _____

(d) $F \cap V \subseteq V$ _____

(e) $F \subseteq V$ _____

(f) $F \cup V \subseteq F$ _____

(g) $F \subseteq F \cap V$ _____

(h) $F \subseteq U$ _____

10. Let $S = \{1, 2, 3\}$.

(a) List every subset of $S$ in the table, sorted by the number of elements.

| Number of elements | Subsets of $S$ | How many? |
|---|---|---|
| 0 | | |
| 1 | | |
| 2 | | |
| 3 | | |

(b) How many subsets does $S$ have in all?

(c) Explain why $\emptyset$ and $S$ itself are both subsets of $S$.

The symbols $\in$ and $\subseteq$ mean different things. In $x \in A$, the thing on the left is an *element*: $x \in A$ says that $x$ is one of the elements of $A$. In $A \subseteq B$, the thing on the left is a *set*: $A \subseteq B$ says that every element of $A$ is also an element of $B$. The symbol '$\in$' is read "is a member of" or "is in" and the symbol '$\subseteq$' is read "is a subset of" or "is contained in."

11. In parts (a)–(j), let $N = \{2, 7, 9\}$. In parts (k)–(p), use the clinic sets $U$, $F$, and $V$.

For parts (a)–(j), write *true* or *false*.

(a) $7 \in N$ _____

(b) $\{7\} \subseteq N$ _____

(c) $\{7\} \in N$ _____

(d) $7 \subseteq N$ _____

(e) $\{2, 9\} \subseteq N$ _____

(f) $\{2, 9\} \in N$ _____

(g) $\emptyset \subseteq N$ _____

(h) $\emptyset \in N$ _____

(i) $N \subseteq N$ _____

(j) $N \in N$ _____

For parts (k)–(p), fill in each blank with $\in$ or $\subseteq$.

(k) C _____ $F$

(l) {C} _____ $F$

(m) {C, I} _____ $V$

(n) I _____ $F \cap V$

(o) $F \cap V$ _____ $F$

(p) H _____ $U$

12. Let $A$ and $B$ be any two sets.

(a) Explain why $A \subseteq A \cup B$ and $B \subseteq A \cup B$.

(b) Explain why $A \cap B \subseteq A$ and $A \cap B \subseteq B$.

(c) In a Venn diagram, how can you see that $A \cap B \subseteq A$ and $A \subseteq A \cup B$?

(d) Explain why $A \cup \emptyset = A$ and $A \cap \emptyset = \emptyset$.

## Complements

If $A$ is a subset of a universal set $U$, then the **complement** of $A$, denoted $A^c$, is the set of elements of $U$ that are not in $A$. The complement of a set is a set, so when we write its elements, we use braces.

13. Use the clinic sets $U$, $F$, and $V$. Find each set.

(a) $F^c =$ _____

(b) $V^c =$ _____

(c) $(F \cup V)^c =$ _____

(d) $(F \cap V)^c =$ _____

(e) $F \cap V^c =$ _____

(f) Describe the sets in parts (c) and (e) in words.

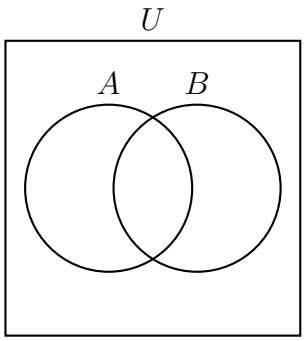
14. Use the clinic sets $U$, $F$, and $V$.

(a) Find $U^{\mathsf{c}}$.

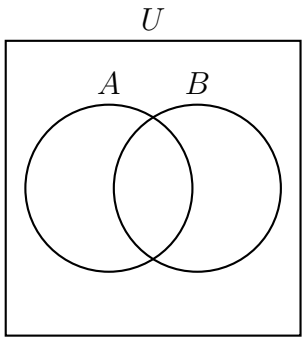
(b) Find $\emptyset^{\mathsf{c}}$.

(c) Now let $U$ be any universal set. Explain why $U^{\mathsf{c}} = \emptyset$ and $\emptyset^{\mathsf{c}} = U$.

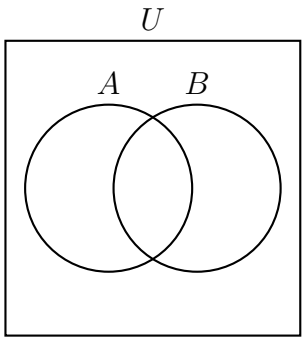
15. In each Venn diagram, the square represents a universal set $U$, and the circles represent subsets $A$ and $B$ of $U$. Shade the region that represents the given set.



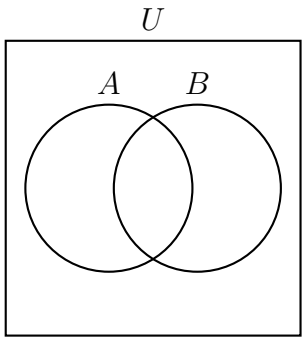

(a) $A \cup B$




(b) $A \cap B$




(c) $A^{\mathrm{c}}$




(d) $B \cap A^{\mathrm{c}}$

## Set or Number? Round Two

16. Use the clinic sets $U$, $F$, and $V$. For each expression, write *set* or *number*, and then find it.

| Expression | Set or number? | What is it? |
|---|---|---|
| $F \cap V$ | | |
| $\lvert F \cap V \rvert$ | | |
| $F^c$ | | |
| $\lvert U \rvert - \lvert F \rvert$ | | |
| $\lvert F \cup V \rvert$ | | |
| $U$ | | |
| $\lvert U \rvert$ | | |
| the patients who have a fever | | |
| the number of patients who have a fever | | |

17. Use the clinic sets $U$, $F$, and $V$. A student gives the answers below. Explain what is wrong with each answer, and write the correct answer.

(a) $F \cap V = 2$

(b) $|F| = \{\text{A}, \text{C}, \text{D}, \text{I}\}$

(c) $F \cup V = \{\text{A}, \text{C}, \text{D}, \text{I}, \text{B}, \text{C}, \text{E}, \text{I}, \text{J}\}$

(d) $F^{\text{c}} = \{\text{B}, \text{E}\}$

(e) $F^{\text{c}} = 4$

(f) $F \cap V = \{\text{A}, \text{B}, \text{C}, \text{D}, \text{E}, \text{I}, \text{J}\}$

## Disjoint Sets

Sets $A$ and $B$ are **disjoint** if $A \cap B = \emptyset$. In other words, disjoint sets have no elements in common: in a Venn diagram, they do not overlap.

18. Use the clinic sets $U$, $F$, and $V$. Ben, Eva, and Hugo go home before noon. They form the set $G = \{\mathrm{B}, \mathrm{E}, \mathrm{H}\}$.

    (a) Find $F \cap G$. Are $F$ and $G$ disjoint?

    (b) Find $V \cap G$. Are $V$ and $G$ disjoint?

19. Use the clinic sets $U$, $F$, and $V$.

    (a) Find $F \cap \emptyset$. Are $F$ and $\emptyset$ disjoint?

    (b) Let $A$ be any set. Explain why $A$ and $\emptyset$ are disjoint.

## The Addition Principle

**The Addition Principle:** If $A$ and $B$ are disjoint sets, then

$$|A \cup B| = |A| + |B|.$$

Since $A \cap B = \emptyset$, no element is in both sets, so adding $|A|$ and $|B|$ counts each element of $A \cup B$ exactly once. The same is true for three or more sets, as long as no two of them have elements in common.

20. Use the clinic sets $U$, $F$, $V$, and $G$.

(a) Find $F \cup G$ by listing its elements, and find $|F \cup G|$.

(b) Find $|F| + |G|$. Explain why your answer is the same as $|F \cup G|$.

(c) The sets $F$ and $F^c$ are disjoint, and $F \cup F^c = U$. Use the Addition Principle to find $|U|$ from $|F|$ and $|F^c|$.

(d) Can you use the Addition Principle to find $|F \cup V|$? Explain.

21. A hospital floor has three wards. Ward $1$ has $12$ patients, Ward $2$ has $9$ patients, and Ward $3$ has $6$ patients. Each patient is in only one ward. Let $W_1$, $W_2$, and $W_3$ be the sets of patients in Wards $1$, $2$, and $3$.

(a) How many patients are in Ward $1$ or Ward $2$? Write your answer using the sets $W_1$ and $W_2$.

(b) How many patients are on the floor?

## The Inclusion-Exclusion Principle

**The Inclusion-Exclusion Principle:** For any two sets $A$ and $B$,

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

Adding $|A|$ and $|B|$ counts each element of $A \cap B$ twice, once in $A$ and once in $B$. Subtracting $|A \cap B|$ makes up for this, so each element of $A \cup B$ is counted exactly once.

22. Use the clinic sets $U$, $F$, and $V$.

(a) Find $|F|$, $|V|$, and $|F \cap V|$.
$|F| =$ _____ $|V| =$ _____ $|F \cap V| =$ _____

(b) Find $|F| + |V|$. Which patients are counted twice?

(c) Use the Inclusion-Exclusion Principle to find $|F \cup V|$. Check your answer by counting the elements of $F \cup V$ in Problem 7(a).

(d) Use the Inclusion-Exclusion Principle to find $|F \cup G|$, where $G = \{\text{B}, \text{E}, \text{H}\}$. Why do you get the same answer as with the Addition Principle in Problem 20?

23. In a clinic's waiting room, $14$ patients are there for a flu shot, $9$ are there for a blood test, and $5$ are there for both. Let $S$ be the set of patients there for a flu shot and $T$ the set of patients there for a blood test.

(a) Write $|S|$, $|T|$, and $|S \cap T|$.

$|S| =$ _____ $\quad |T| =$ _____ $\quad |S \cap T| =$ _____

(b) How many patients are there for a flu shot or a blood test (or both)?

(c) Explain why the answer to part (b) is not $14 + 9 = 23$.

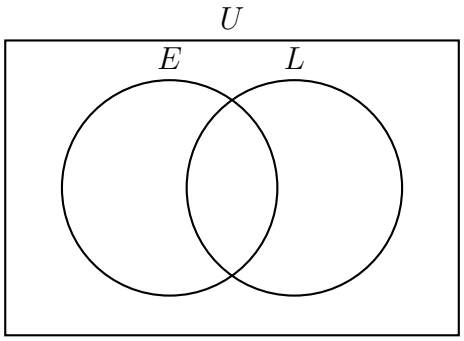
## Putting It Together

24. Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ be the universal set. Let $E$ be the set of even numbers in $U$, and let $L$ be the set of numbers in $U$ that are less than $5$.

(a) List the elements of $E$ and $L$.

$E =$ _____ $L =$ _____

(b) Write each number of $U$ in the correct region of the Venn diagram.



(c) Find each set.

$E \cup L =$ _____

$E \cap L =$ _____

$E^c =$ _____

$(E \cup L)^c =$ _____

(d) Find $|E \cup L|$ by counting the elements of $E \cup L$. Then check your answer using the Inclusion-Exclusion Principle.

(e) $|L| = 4$, and $4$ is also an element of $L$. Does this mean that $L$ and $|L|$ are the same thing? Explain.